3. Read IMU Data

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1. Camera Introduction

The D-ROBOTICS binocular IMU camera comes with complete calibration parameters, including binocular calibration parameters, IMU intrinsic parameters, and extrinsic parameters between the binocular camera and IMU. Users can use it directly without additional calibration. Using these parameters, high-precision depth maps can be computed through binocular stereo matching algorithms, enabling real-time 3D environment perception. Meanwhile, the camera data can be applied to open-source Visual-Inertial Odometry (VIO) algorithms such as OpenVINS, which can compute camera pose and trajectory information. Suitable for robot navigation, obstacle avoidance and other applications, providing developers with plug-and-play depth perception and visual-inertial fusion capabilities.

MIPI camera code repository: https://github.com/D-Robotics/hobot_mipi_cam

2. Supported Platforms

Platform	System Support	Example Functionality
RDK X5, RDK X5 Module	Ubuntu 22.04 (Humble)	Start binocular camera, output binocular images and IMU data

3. Preparation

3.1. RDK Platform

1. RDK has been flashed with RDK OS system
2. RDK has successfully installed TogetheROS.Bot
3. Confirm that the PC can access RDK through the network

3.2. System and Package Versions

	Version	Query Method
RDK X5 system image version	3.4.1 or above	`cat /etc/version`
tros-humble-hobot-stereonet package	2.5.0 or above	`apt list | grep tros-humble-hobot-stereonet/`
tros-humble-mipi-cam package	2.5.0 or above	`apt list | grep tros-humble-mipi-cam/`

- If the system image version doesn't meet requirements, please refer to the corresponding section for image flashing
- If the package version doesn't meet requirements, execute the following commands to update:

```
sudo apt update
sudo apt install --only-upgrade tros-humble-hobot-stereonet
sudo apt install --only-upgrade tros-humble-mipi-cam
```

- If the above commands cannot update to the latest version, you need to modify the apt source file to beta source:

```
# Modify to beta source, execute the following command:
sudo echo 'deb [signed-by=/usr/share/keyrings/sunrise.gpg] http://archive.d-robotics.cc/ubuntu-rdk-x5-beta  jammy main' | sudo tee /etc/apt/sources.list.d/sunrise.list
apt update

# If you need to switch back to the official source, execute the following command:
sudo echo 'deb [signed-by=/usr/share/keyrings/sunrise.gpg] http://archive.d-robotics.cc/ubuntu-rdk-x5  jammy main' | sudo tee /etc/apt/sources.list.d/sunrise.list
apt update
```

4. Camera Connection and Preparation

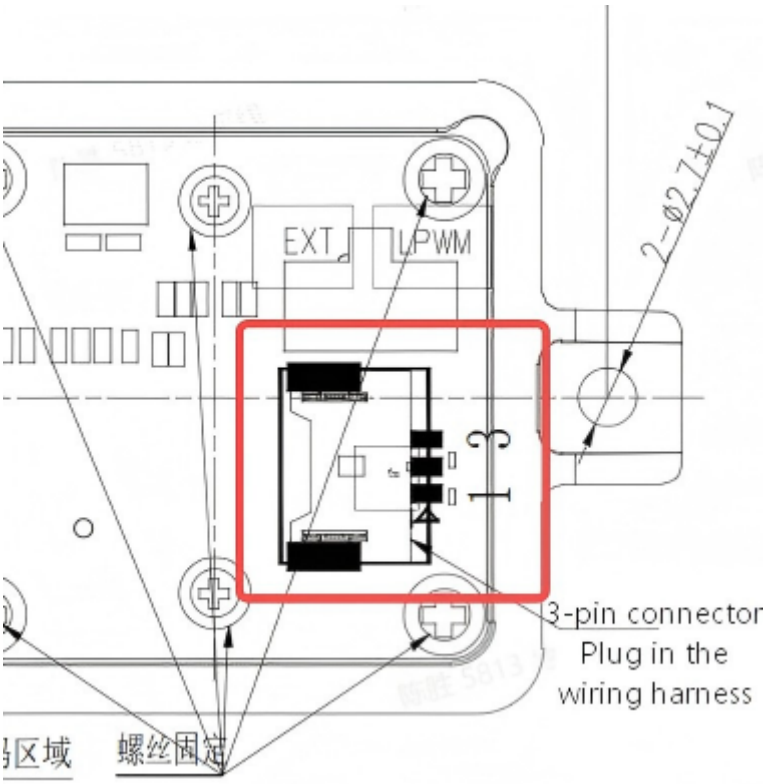
4.1. Binocular IMU Camera

- Note that the camera has two versions: black metal shell and acrylic shell, both with the same functionality (the cable may vary based on what you receive)

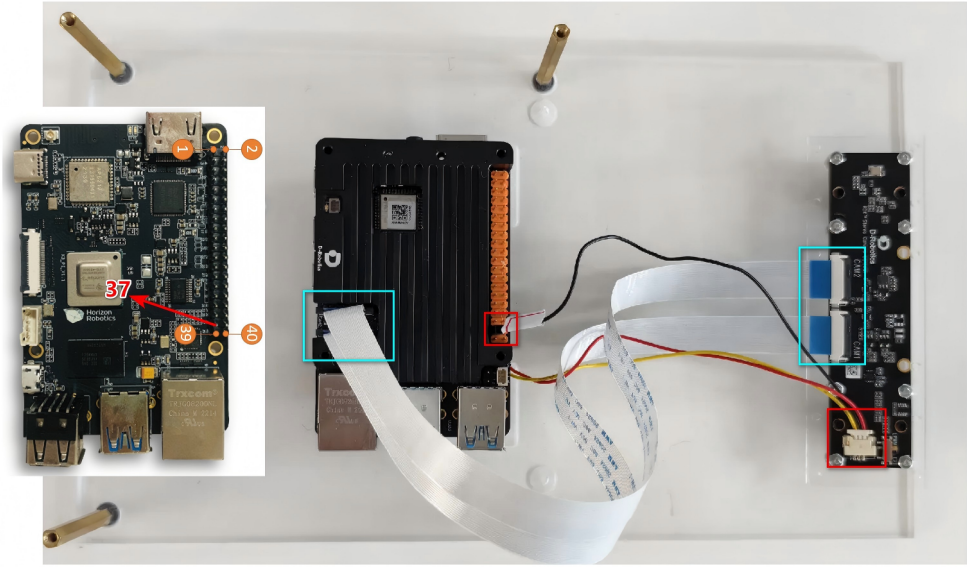
4.2. Hardware Connection

1. Set the switch behind the module to `EXT` mode, not `LPWM` mode
2. No other pins need to be connected except the black Dupont wire connecting to RDK X5's `37` pin, which is needed for external trigger control of camera and IMU time synchronization
3. Camera 3Pin pin definition

PIN	Pin Number	Definition	Description	Type
Pin	1	INT1	Input	IMU interrupt signal 1
Pin	2	INT2	Input/Output	IMU interrupt signal 2; active when the switch is set to "EXT"
Pin	3	DGND	GND	Digital GND



4. For the motherboard pin definition, please refer to [3.1.1 Pin Definition and Application]



RDX X5 Board 40Pin Table																
Reuse function3	Reuse function2	Reuse function1	Reuse function 0	Function Description	X5 Pin Number	BCM Encoding	CVM Function	Physical Pin Board Encoding		CVM Function	BCM Encoding	X5 Pin Number	Function Description	Reuse function 0	Reuse function1	Reuse function2	Reuse function3
				3.3V power signal			VDD_3V3	1	2	VDD_5V			5V power signal				
	LSIO_GPIO0_11	SDA5	UART3_TXD	I2C0 data signal	387	2	I2C5_SDA	3	4	VDD_5V			5V power signal				
	LSIO_GPIO0_10	SCL5	UART3_RXD	I2C0 clock signal	389	3	I2C5_SCL	5	6	GND			GND signal				
		DSP_GPIO_09	DSP_MCLK1	I2S0 MCLK clock signal	420	4	I2S1_MCLK	7	8	UART_TXD	14	383	UART1 send signal	UART1_TXD		LSIO_GPIO0_5	
				GND signal			GND	9	10	UART_RXD	15	384	UART1 receive signal	UART1_RXD		LSIO_GPIO0_4	
	LSIO_GPIO0_1		UART7_TXD	GPIO17 signal	380	17	GPIO17	11	12	I2S1_BCLK	18	421	I2S1 BCLK clock signal	I2S1_BCLK	DSP_GPIO_10		
	LSIO_GPIO0_0		UART7_RXD	GPIO27 signal	379	27	GPIO27	13	14	GND			GND signal				
	LSIO_GPIO0_9		UART2_TXD	GPIO22 signal	388	22	GPIO22	15	16	GPIO23	23	382	GPIO23 signal	UART0_RTS	UART6_TXD	LSIO_GPIO0_3	
				3.3V power signal			VDD_3V3	17	18	GPIO24	24	402	GPIO24 signal	SPI2_MOSI	LSIO_GPIO0_23		LSIO_PWM_OUT3
	JTG_TDO	LSIO_GPIO0_19	SPI1_MOSI	SPI1 MOSI signal	398	10	SPI1_MOSI	19	20	GND			GND signal				
	JTG_TDI	LSIO_GPIO0_18	SPI1_MISO	SPI1 MISO signal	397	9	SPI1_MISO	21	22	GPIO25	25	387	GPIO25 signal	UART2_RXD		LSIO_GPIO0_8	
	JTG_TCK	LSIO_GPIO0_16	SPI1_SCLK	SPI1 CLK signal	395	11	SPI1_SCLK	23	24	SPI_CSN0	8	394	SPI1 SSN1 signal	SPI1_CSN1	LSIO_GPIO0_15	JTG_TMS	
				GND signal			GND	25	26	SPI_CSN1	7	396	SPI1 SSN0 signal	SPI1_CSN0	LSIO_GPIO0_17	JTG_TRSTN	
LSIO_PWM_OUT5		LSIO_GPIO1_8	SDA0	I2C3 clock signal	355	0	I2C0_SDA	27	28	I2C0_SCL	1	354	I2C0 signal	SCL0	LSIO_GPIO1_7		LSIO_PWM_OUT4
LSIO_PWM_OUT0		LSIO_GPIO0_20	SPI2_SCLK	GPIO5 signal	399	5	GPIO5	29	30	GND			GND signal				
LSIO_PWM_OUT1	TIME_SYNC2	LSIO_GPIO0_21	SPI2_SSN	GPIO6 signal	400	6	GPIO6	31	32	PWM6	12	356	PWM6 signal	SCL1	LSIO_GPIO1_9	TIME_SYNC1	LSIO_PWM_OUT6
LSIO_PWM_OUT7		LSIO_GPIO1_10	SDA1	PWM0 signal	357	13	PWM7	33	34	GND			GND signal				
		DSP_GPIO_11	I2S1_LRCK	I2S0 LRCK signal	422	19	I2S1_LRCK	35	36	GPIO16	16	381		UART0_CTS	UART6_RXD	LSIO_GPIO0_2	
LSIO_PWM_OUT2		LSIO_GPIO0_22	SPI2_MISO	GPIO26 signal	401	26	GPIO26	37	38	I2S1_SDIN	20	423	I2S1 DI signal	I2S1_DIN	DSP_GPIO_12		
				GND signal			GND	39	40	I2S1_SDOUT	21	424	I2S1 DO signal	I2S1_DOUT	DSP_GPIO_13		

4.3. RDK X5 Configuration

1. RDK X5 needs to be configured to read IMU data. First check if the system version meets the requirements, need `3.4.1` or above:

```
cat /etc/version
```

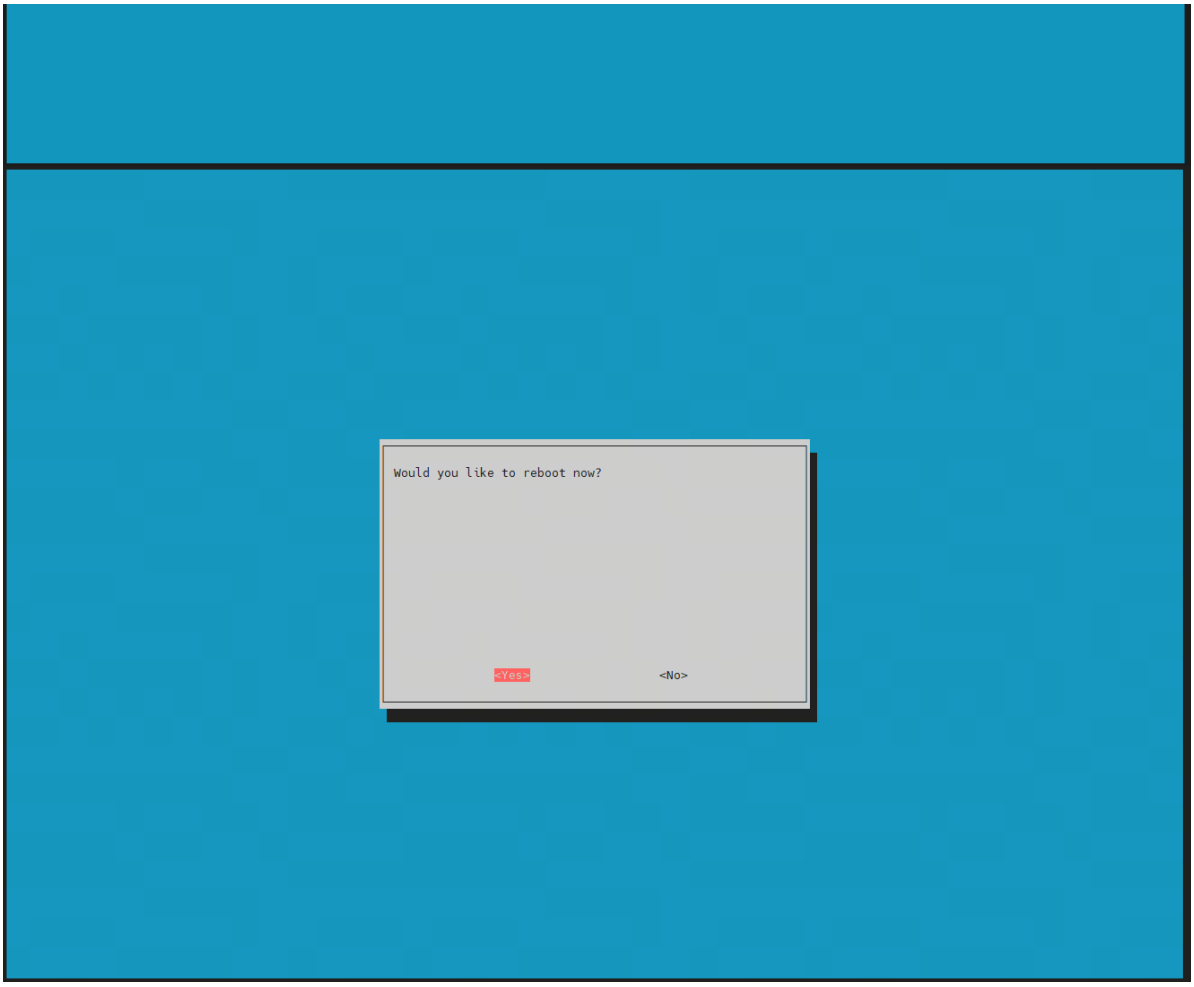
2. Run the following command on RDK X5 for configuration:

```
srpi-config
```

```
D-Robotics RDK X5 V1.0

┤ RDK Software Configuration Tool (srpi-config) ├

1 System Options        Configure system settings
2 Display Options       Configure display settings
3 Interface Options     Configure connections to peripherals
4 Performance Options   Configure performance settings
5 Localisation Options  Configure language and regional settings
6 Advanced Options      Configure advanced settings
7 Sensor Profiles       Configure ISP settings for different FOVs
8 Update                Update this tool to the latest version
9 About srpi-config     Information about this configuration tool

<Select>                                    <Finish>
```

```
┤ RDK Software Configuration Tool (srpi-config) ├

I1 SSH                    Enable/disable remote command line access using SSH
I2 VNC                    Enable/disable graphical remote desktop access
I3 Peripheral bus config  Enable/disable peripheral bus(spi,i2c,serial,i2s,etc.)
I4 Configure Wi-Fi antenna Configure Wi-Fi antennas to use onboard or external antenna
I5 Audio                  Configure audio peripheral functions
I6 IMU                    Configure IMU peripheral functions
I7 V4L2                   Configure V4L2 functions

<Select>                                    <Back>
```

```
┤ RDK Software Configuration Tool (srpi-config) ├

BMI088-SPI-Interface      Use SPI1 as the data interface for IMU.
BMI088-I2C-Interface      Use I2C5 as the data interface for IMU.
ICM42688-I2C-Interface    Use I2C4 or I2C6 as the data interface for Stereo Camera IMU.
UNSET                     Unset IMU driver hat

<Select>                                    <Back>
```



3. After reboot, confirm that the directory `/sys/bus/iio/devices/` has `iio:device1` and `iio:device2`, which means the configuration is successful:

```
ll /sys/bus/iio/devices/
```

5. Read IMU Data

5.1 Execute the following commands to start the camera

Switch to root user, username: root, password: root

```
source /opt/tros/humble/setup.bash
```

```
ros2 launch mipi_cam mipi_cam_dual_channel.launch.py \
mipi_channel:=2 mipi_channel2:=0 \
mipi_lpwm_enable:=True mipi_frame_ts_type:=realtime \
mipi_image_width:=816 mipi_image_height:=960 \
mipi_image_framerate:=10.0 mipi_gdc_enable:=True \
mipi_out_format:=nv12 \
log_level:=info
```

Parameter explanation:

- mipi_channel:=2 mipi_channel2:=0 Adjust left-right camera stitching order
- mipi_lpwm_enable:=True Enable LPWM hardware synchronization
- mipi_frame_ts_type:=realtime Timestamp uses system time
- mipi_image_width:=816 mipi_image_height:=960 Adjust image resolution, maximum can be set to 1088*1280
- mipi_image_framerate:=10.0 Adjust camera frame rate, maximum can be adjusted to 30.0
- mipi_gdc_enable:=True Enable GDC correction, will publish corrected binocular images, otherwise will publish images with distortion
- mipi_out_format:=nv12 Set image format, supports nv12/bgr8
- log_level:=info Log level, info level can print calibration parameters, if you don't need to display that much information, set to warn

5.2 View IMU Data

View camera topic information

```
ros2 topic list -v
```

```
root@ubuntu:~# ros2 topic list -v
Published topics:
 * /image_combine_raw [sensor_msgs/msg/Image] 1 publisher
 * /image_combine_raw/left/camera_info [sensor_msgs/msg/CameraInfo] 1 publisher
 * /image_combine_raw/right/camera_info [sensor_msgs/msg/CameraInfo] 1 publisher
 * /image_left_raw [sensor_msgs/msg/Image] 1 publisher
 * /image_left_raw/camera_info [sensor_msgs/msg/CameraInfo] 1 publisher
 * /image_right_raw [sensor_msgs/msg/Image] 1 publisher
 * /image_right_raw/camera_info [sensor_msgs/msg/CameraInfo] 1 publisher
 * /imu_data [sensor_msgs/msg/Imu] 1 publisher
 * /parameter_events [rcl_interfaces/msg/ParameterEvent] 2 publishers
 * /rosout [rcl_interfaces/msg/Log] 2 publishers

Subscribed topics:
 * /parameter_events [rcl_interfaces/msg/ParameterEvent] 1 subscriber
```

- /image_combine_raw is the up-down stitched binocular image, mipi_channel:=2 mipi_channel2:=0 parameter controls the stitching order
- /image_left_raw /image_right_raw are left-right camera data topics, mipi_channel:=2 mipi_channel2:=0 parameter controls the order
- /imu_data is the IMU data topic, publishing gyroscope and accelerometer data

View IMU topic data

```
ros2 topic echo
```

You can see the change of IMU data by changing the camera's posture.

Note: In the IMU data topic, `angular_velocity` unit is rad/s, `linear_acceleration` unit is m/s², the gravity acceleration value is `9.81`

```
root@ubuntu:~# ros2 topic echo /imu_data
header:
  stamp:
    sec: 1774582872
    nanosec: 165800960
  frame_id: imu_link
orientation:
  x: 0.0
  y: 0.0
  z: 0.0
  w: 1.0
orientation_covariance:
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
angular_velocity:
  x: -0.002130528911948204
  y: -0.006391586735844612
  z: -0.001065264455974102
angular_velocity_covariance:
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
linear_acceleration:
  x: -0.53648442029953
  y: 0.3400927782058716
  z: -9.398056983947754
linear_acceleration_covariance:
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
```